SCIENCE MASTER X

Complete Academic Study Notes • Class 10 Science

CHAPTER 13

Our Environment

(Full Chapter)

> Ecosystem & Components • Food Chains & Food Webs • 10% Law of Energy Transfer • Biological Magnification • Ozone Layer Depletion • Waste Management & Biodegradability

Chapter Details

Target Level: CBSE Class 10

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📖 Chapter Overview & Basic Concepts

Environment consists of everything that surrounds us, including physical, chemical, and biological factors. All organisms in an environment interact with each other and with their abiotic surroundings to form a balanced ecosystem.

SECTION 1: ECOSYSTEM AND ITS COMPONENTS

1.1 What is an Ecosystem?

- **Ecosystem:** A self-sustaining structural and functional unit of the biosphere where living organisms interact among themselves and with their non-living physical environment.
- **Natural Ecosystems:** Created by nature without human intervention (e.g., Forests, Lakes, Ponds, Oceans).
- **Artificial (Man-made) Ecosystems:** Created and maintained by humans (e.g., Aquariums, Crop fields, Gardens).

1.2 Components of an Ecosystem

- **Abiotic Components:** Non-living physical and chemical factors like light, temperature, water, soil, air, and inorganic nutrients.
- **Biotic Components:** All living organisms divided based on their mode of nutrition:

Biotic Component	Mode of Nutrition	Role in Ecosystem	Examples
Producers (Autotrophs)	Photosynthesis (Synthesize organic food from inorganic compounds)	Traps solar energy and provides food for all organisms.	Green plants, Blue-green algae
Consumers (Heterotrophs)	Depend on producers directly or indirectly for nutrition	Controls population levels and transfers energy across levels.	Herbivores, Carnivores, Omnivores, Parasites
Decomposers (Saprotrophs)	Break down complex organic substances of dead organisms into simple inorganic nutrients	Recycles nutrients back into the soil and cleans the environment naturally.	Bacteria and Fungi

1.3 Types of Consumers

- **Herbivores (Primary Consumers):** Eat plants directly (e.g., Deer, Cow, Grasshopper).
- **Carnivores (Secondary/Tertiary Consumers):** Eat other animals (e.g., Frog, Snake, Lion).
- **Omnivores:** Eat both plants and animals (e.g., Humans, Crows).
- **Parasites:** Live on or inside host bodies for nutrition without killing them immediately (e.g., *Cuscuta*, Tapeworm).

> **NCERT High-Yield Insight:** An aquarium is an artificial ecosystem that requires regular cleaning because it lacks natural decomposers to break down organic waste, unlike natural ponds or lakes.

SECTION 2: FOOD CHAINS, FOOD WEBS & ENERGY FLOW

2.1 Food Chain

- **Food Chain:** A linear sequence of organisms where energy is transferred in the form of food from one organism to another.
- **Trophic Levels:** Each step or level of a food chain where energy transfer takes place.

Producers (1st Trophic Level - T_1) → **Primary Consumers** (2nd Trophic Level - T_2) → **Secondary Consumers** (3rd Trophic Level - T_3) → **Tertiary Consumers** (4th Trophic Level - T_4)

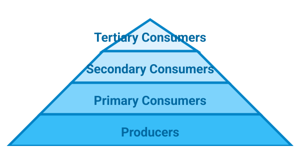
2.2 Energy Flow Laws in Ecosystems

- **Unidirectional Flow:** Energy flows in only one direction (Sun → Producers → Herbivores → Carnivores). Energy captured by autotrophs does not revert back to solar input.
- **1% Solar Capture Law:** Green plants capture only about **1%** of the total solar energy falling on their leaves and convert it into food energy.
- **Lindeman's 10% Energy Law:** Only **10%** of the energy available at a trophic level is transferred to the next higher trophic level. The remaining **90%** is lost to the surroundings as heat and used for life processes (digestion, respiration, growth).

Important Formula / Rule:

$$\text{Energy at Next Trophic Level} = \text{Energy at Current Level} \times 0.10$$

📌 **Diagram: Trophic Levels Pyramid**



2.3 Food Web & Biological Magnification

- **Food Web:** A network of interconnected food chains operating naturally in an ecosystem. Provides stability to the ecosystem.
- **Biological Magnification (Biomagnification):** The progressive accumulation of non-biodegradable toxic chemicals (like pesticides, DDT) at each successive trophic level of a food chain.

- Since **humans** occupy the topmost trophic level in most food chains, the maximum concentration of these harmful chemicals accumulates in human bodies.

SECTION 3: ENVIRONMENTAL PROBLEMS & OZONE LAYER

3.1 Ozone Layer Depletion

- **Ozone (O_3):** A molecule formed by three atoms of oxygen. It is deadly poisonous at ground level, but at higher atmospheric levels (stratosphere), it shields Earth from harmful solar Ultraviolet (UV) radiation.
- **Harmful Effects of UV Rays:** Skin cancer, eye cataracts, weakened immune system, damage to crops and phytoplankton.

3.2 Formation & Reaction Mechanism of Ozone

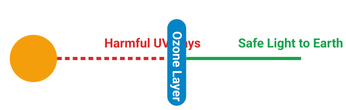
Important Chemical Reactions:

1. High-energy UV radiation splits oxygen gas (O_2) into free oxygen atoms (O): $$\text{O}_2 \xrightarrow{\text{UV Radiation}} \text{O} + \text{O}$$ 2. Free oxygen atom combines with molecular oxygen (O_2) to form ozone (O_3): $$\text{O} + \text{O}_2 \rightarrow \text{O}_3$$

3.3 Depletion Cause & Control Measures

- **Main Cause:** Synthetic chemicals like **Chlorofluorocarbons (CFCs)** used in refrigerators, air conditioners, and fire extinguishers release chlorine radicals that destroy ozone molecules.
- **UNEP Agreement (1987):** The United Nations Environment Programme successfully forged an international agreement to freeze CFC production at 1986 levels.

📌 **Diagram: Ozone Layer Protection & UV Shield**



NCERT High-Yield Insight: Ozone depletion was first prominently observed over Antarctica in the mid-1980s, commonly known as the "Ozone Hole".

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SECTION 4: MANAGEMENT OF GARBAGE & WASTE

4.1 Types of Waste Materials

Property	Biodegradable Waste	Non-biodegradable Waste
Definition	Substances that can be broken down into simpler non-toxic substances by biological action of decomposers.	Substances that cannot be broken down by biological action of microorganisms.
Decomposer Action	Enzymes produced by bacteria and fungi break their chemical bonds.	Enzymes are specific in action and cannot break their synthetic chemical bonds.
Environmental Impact	Persists for short duration; eco-friendly if managed well.	Persists in environment for a long time; causes soil and water pollution.
Examples	Fruit skins, vegetable peels, paper, cow dung, cotton.	Plastics, polythene bags, glass, metals, DDT.

4.2 Waste Disposal Methods

- **Composting / Vermicomposting:** Organic waste converted into nutrient-rich manure using earthworms.
- **Recycling:** Processing plastic, glass, and paper to create new usable items.
- **Incineration:** Burning waste at high temperatures to reduce volume (used for medical waste).
- **Sewage Treatment Plants (STP):** Treating waste water before releasing it into water bodies.

❓ Important Practice Questions

Q1: Why are crop fields and aquariums considered artificial ecosystems while lakes are natural?
Ans: Crop fields and aquariums are created, structured, and maintained by human intervention. Lakes are naturally occurring, self-sustaining ecosystems managed by nature.

Q2: Calculate the amount of energy available to lions if 20,000 J of solar energy falls on green plants.
Ans: Plants capture 1% of solar energy $= 20,000 \times 0.01 = 200\text{ J}$ (Producers).
Herbivores get 10% of 200 J $= 20\text{ J}$.
Lions (Carnivores) get 10% of 20 J $= \mathbf{2\text{ J}}$.

Q3: What is Biological Magnification? Which organism gets the highest concentration of toxic chemicals?
Ans: The progressive accumulation of non-biodegradable toxic chemicals at successive trophic levels. The organism at the highest trophic level (e.g., humans) accumulates the maximum concentration.

Q4: Write the chemical equation for ozone formation in the upper atmosphere. Why is CFC harmful?
Ans: Equations: $\text{O}_2 \xrightarrow{\text{UV}} \text{O} + \text{O}$ and $\text{O} + \text{O}_2 \rightarrow \text{O}_3$. CFCs release active chlorine atoms under UV radiation which destroy ozone

molecules in the stratosphere.